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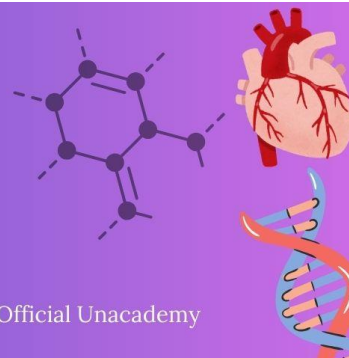
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LAST 10 YEARS

NEET PYQ



Chapter 13 : Plant Growth and Development

1. Read the following statements on plant growth and development.

[NEET-2025]

- (A) Parthenocarpy can be induced by auxins.
- (B) Plant growth regulators can be involved in promotion as well as inhibition of growth.
- (C) Dedifferentiation is a pre-requisite for re-differentiation.
- (D) Absciscic acid is a plant growth promoter.
- (E) Apical dominance promotes the growth of lateral buds.

Choose the option with all correct statements.

- (1) A, B, C only
 - (2) A, C, E only
 - (3) A, D, E only
 - (4) B, D, E only
2. Which one of the following phytohormones promotes nutrient mobilization which helps in the delay of leaf senescence in plants?

[NEET-2025]

- (1) Ethylene
- (2) Absciscic acid
- (3) Gibberellin
- (4) Cytokinin

3. Match List-I with List-II :

[Re-NEET-2024]

List-I

List-II

- | | |
|-------------------|---|
| A. Absciscic acid | I. Promotes female flowers in cucumber |
| B. Ethylene | II. Helps seeds to withstand desiccation |
| C. Gibberellin | III. Helps in nutrient mobilisation |
| D. Cytokinin | IV. Promotes bolting in beet, cabbage etc |

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-I, C-III, D-IV

4. F. Skoog observed that callus proliferated from the internodal segments of tobacco stem when auxins was supplied with one of the following except : **[Re-NEET-2024]**

- | | |
|---------------------------------|-------------------|
| (1) Extract of Vascular tissues | (2) Coconut milk |
| (3) Absciscic acid | (4) Yeast Extract |

5. Given below are some statements about plant growth regulators. **[Re-NEET-2024]**

- A. All Gas are acidic in nature.
B. Auxins are antagonists to Gas.
C. Zeatin was isolated from coconut milk.
D. Ethylene induces flowering in Mango.
E. Absciscic acid induces parthenocarpy.

Choose the correct set of statements from the option given below:

- | | |
|-------------|-------------|
| (1) A, C, D | (2) B, E |
| (3) A, B, C | (4) B, D, E |

6. Auxin is used by gardeners to prepare weed-free lawns. But no damage is caused to grass as auxin

- (1) promotes apical dominance. **[NEET-2024]**
(2) promotes abscission of mature leaves only.
(3) does not affect mature monocotyledonous plants.
(4) can help in cell division in grasses, to produce growth

7. Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield? **[NEET-2024]**

- | | |
|---------------|--------------------|
| (1) Auxin | (2) Gibberellin |
| (3) Cytokinin | (4) Absciscic acid |

8. The phenomenon which is influenced by auxin and also played a major role in its discovery :

- | | | |
|------------------|----------------------|----------------------------|
| (1) Gravitropism | (2) Apical Dominance | [NEET-2023-MANIPUR] |
| (3) Phototropism | (4) Root initiation | |

9. Match List-I with List-II **[NEET-2023-MANIPUR]**

List-I

List-II

- | | |
|-----------------|--|
| (A) Auxin | (I) Promotes female flower formation in cucumber |
| (B) Gibberellin | (II) Overcoming apical dominance |
| (C) Cytokinin | (III) Increase in the length of grape stalks |
| (D) Ethylene | (IV) Promotes flowering in pineapple |

Choose the correct answer from the options given below:

- | | |
|--|--|
| (1) (A)-(I), (B)-(III), (C)-(IV), (D)-(II) | (2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV) |
| (3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III) | (4) (A)-(IV), (B)-(III), (C)-(II), (D)-(I) |

10. Which of the following statements is not correct? **[NEET-2023-MANIPUR]**
- (1) Plant growth is generally determinate.
 - (2) Plant growth is measurable.
 - (3) Phase of cell elongation of plant cells is characterized by increased vacuolation.
 - (4) Cells in the meristematic phase of growth exhibit abundant plasmodesmatal connections.
11. Which hormone promotes internode/petiole elongation in deep water rice? **[NEET-2023]**
- (1) GA3
 - (2) Kinetin
 - (3) Ethylene
 - (4) 2, 4-D
12. Spraying of which of the following phytohormone on juvenile conifers helps hastening the maturity period, that leads early seed production? **[NEET-2023]**
- (1) Indole-3-butyric Acid
 - (2) Gibberellic Acid
 - (3) Zeatin
 - (4) Absciscic Acid
13. In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as **[NEET-2023]**
- (1) Differentiation
 - (2) Dedifferentiation
 - (3) Development
 - (4) Senescence
14. Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants: **[NEET-2022]**
- (1) Gibberellin
 - (2) Ethylene
 - (3) Cytokinin
 - (4) ABA
15. Which one of the following plants does not show plasticity ? **[NEET-2022]**
- (1) Coriander
 - (2) Buttercup
 - (3) Maize
 - (4) Cotton

16. The gaseous plant growth regulator is used in plants to : **[NEET-2022]**
- (1) promote root growth and root hair formation to increase the absorption surface
 - (2) help overcome apical dominance
 - (3) kill dicotyledonous weeds in the fields
 - (4) speed up the malting process
17. Which of the following growth regulators is an adenine derivative ? **[Re-NEET-2022]**
- (1) Auxin
 - (2) Cytokinin
 - (3) Ethylene
 - (4) Absciscic acid
18. The phenomenon by which the undividing parenchyma cells start to divide mitotically during plant tissue culture is called as : **[Re-NEET-2022]**
- (1) Differentiation
 - (2) Dedifferentiation
 - (3) Redifferentiation
 - (4) Secondary growth
19. The ability of plants to follow different pathways in response to environment leading to formation of different kinds of structures is called : **[Re-NEET-2022]**
- (1) Redifferentiation
 - (2) Development
 - (3) Plasticity
 - (4) Differentiation
20. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called : **[NEET-2021]**
- (1) Elasticity
 - (2) Flexibility
 - (3) Plasticity
 - (4) Maturity
21. The site of perception of light in plants during photoperiodism is : **[NEET-2021]**
- (1) Shoot apex
 - (2) Stem
 - (3) Axillary bud
 - (4) Leaf
22. The plant hormone used to destroy weeds in a field is : **[NEET-2021]**
- (1) IAA
 - (2) NAA
 - (3) 2,4-D
 - (4) IBA

23. Which of the following is incorrectly matched pair? [NEET-2021]

- (1) Ethylene – Cousins
- (2) Kinetin – Skoog and Miller
- (3) Gibberellic acid – F.W. Went
- (4) Auxin – Darwin and Darwin

24. Match the columns I and II, and choose the correct combination from the options given. [NEET-2021]

Column-I	Column II
a. Zeatin	i. Tracheary element
b. Differentiation	ii. Secondary xylem
c. Auxin	iii. Cytokinin
d. Redifferentiation	iv. Indole

- (1) a–iii, b–i, c–ii, d–iv
- (2) a–iii, b–ii, c–i, d–iv
- (3) a–iii, b–i, c–iv, d–ii
- (4) a–iv, b–iii, c–ii, d–i

25. The process of growth is maximum during : [NEET-2020]

- | | |
|---------------|----------------|
| (1) Dormancy | (2) Log phase |
| (3) Lag phase | (4) Senescence |

26. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop. [NEET-2020]

- (1) Absciscic acid
- (2) Cytokinin
- (3) Gibberellin
- (4) Ethylene

27. Inhibitory substances in dormant seeds cannot be removed by subjecting seeds to :

- (1) Gibberellic acid [NEET-2020 COVID-19]
- (2) Nitrate
- (3) Ascorbic acid
- (4) Chilling conditions

28. Match the following concerning the activity/function and the phytohormone involved :-

- | | | |
|--------------------|--------------------|----------------------|
| (a) Fruit ripener | (i) Absciscic acid | [NEET-2020 COVID-19] |
| (b) Herbicide | (ii) GA3 | |
| (c) Bolting agent | (iii) 2, 4-D | |
| (d) Stress hormone | (iv) Ethephon | |
- Select the correct option from following :-

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

29. What is the site of perception of photoperiod necessary for induction of flowering in plants? **[NEET-2019]**

(1) Lateral buds

(2) Pulvinus

(3) Shoot apex

(4) Leaves

30. It takes very long time for pineapple plants to produce flowers. Which combination of hormones can be applied to artificially induce flowering in pineapple plants throughout the year to increase yield?

(1) Auxin and Ethylene

[NEET-2019]

(2) Gibberellin and Cytokinin

(3) Gibberellin and Absciscic acid

(4) Cytokinin and Absciscic acid

31. Removal of shoot tips is a very useful technique to boost the production of tea leaves. This is because: -

(1) Gibberellins prevent bolting and are inactivated

[NEET-2019-ODISHA]

(2) Auxins prevent leaf drop at early stages

(3) Effect of auxins is removed and growth of lateral buds is enhanced.

(4) Gibberellins delay senescence of leaves.

32. In order to increase the yield of sugarcane crop, which of the following plant growth regulators should be sprayed?

[NEET-2019-ODISHA]

(1) Ethylene

(2) Auxins

(3) Gibberellins

(4) Cytokinins

33. Fruit and leaf drop at early stages can be prevented by the application of:

[NEET-2017]

(1) Ethylene

(2) Auxins

(3) Gibberellic acid

(4) Cytokinins

34. The Avena curvature is used for bioassay of:

[NEET- I 2016]

(1) ABA

(2) GA3

(3) IAA

(4) Ethylene

35. You are given a tissue with its potential for differentiation in an artificial culture. Which of the following pairs of hormones would you add to the medium to secure shoots as well as roots? **[NEET- II 2016]**

- (1) Auxin and abscisic acid
- (2) Gibberellin and abscisic acid
- (3) IAA and gibberellin
- (4) Auxin and cytokinin

36. Phytochrome is a: **[NEET- II 2016]**

- (1) Lipoprotein
- (2) Chromoprotein
- (3) Flavoprotein
- (4) Glycoprotein

ANSWER KEY

1. (1)
2. (4)
3. (3)
4. (3)
5. (1)
6. (3)
7. (2)
8. (3)
9. (4)
10. (1)
11. (3)
12. (2)
13. (2)
14. (2)
15. (3)
16. (1)
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26. (3)
27. (3)
28. (3)
29. (4)
30. (1)
31. (3)
32. (3)
33. (2)
34. (3)
35. (4)
36. (2)